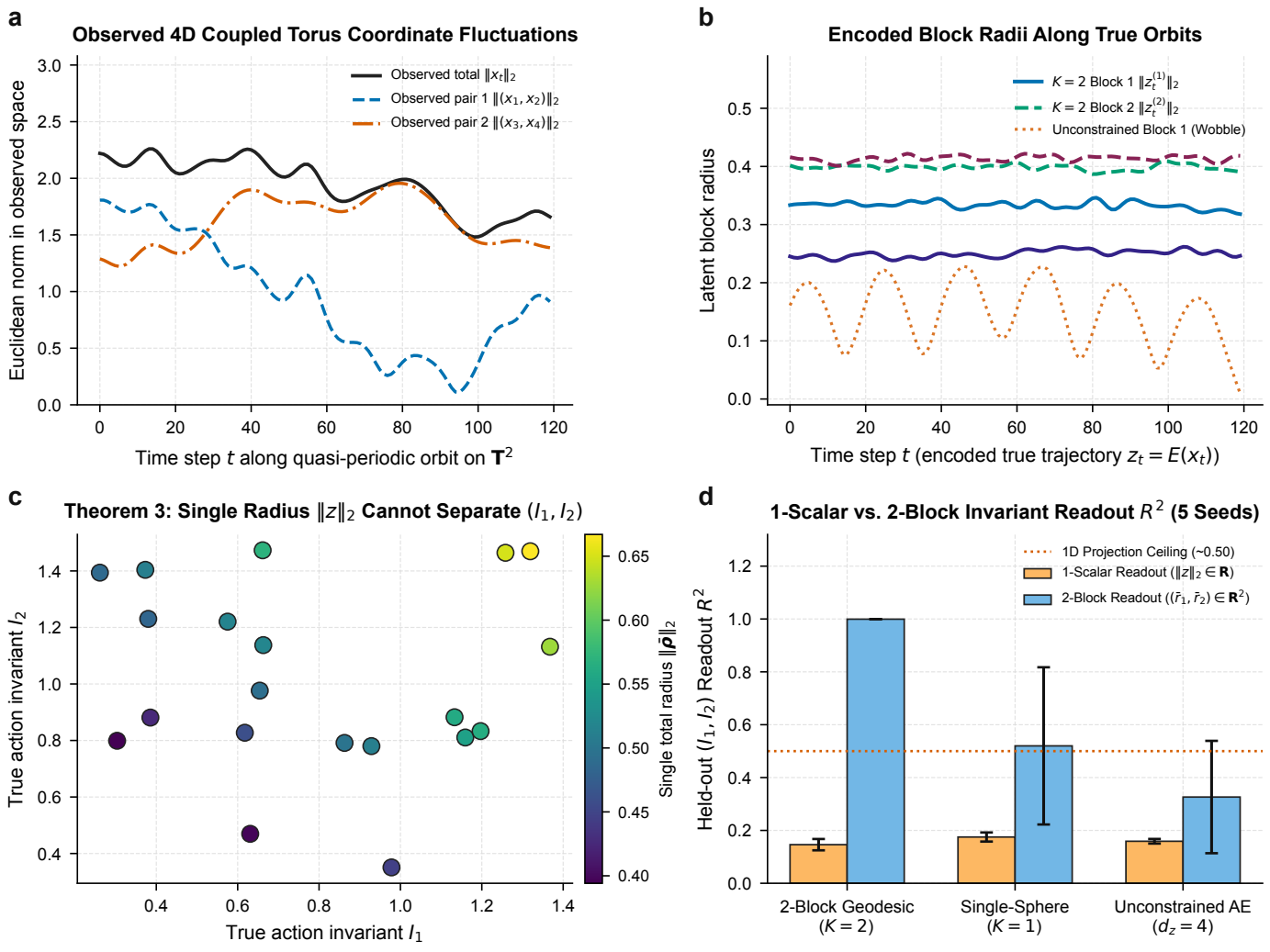

a
Observed 4D Coupled Torus Coordinate Fluctuations
Observed total $\|x_t\|_2$
Observed pair 1 $\|(x_1, x_2)\|_2$
Observed pair 2 $\|(x_3, x_4)\|_2$
Euclidean norm in observed space
Time step t along quasi-periodic orbit on $\mathbf{T}^2$
b
Encoded Block Radii Along True Orbits
$K = 2$ Block 1 $\|z_t^{(1)}\|_2$
$K = 2$ Block 2 $\|z_t^{(2)}\|_2$
Unconstrained Block 1 (Wobble)
Latent block radius
Time step t (encoded true trajectory $z_t = E(x_t)$)
c
Theorem 3: Single Radius $\|z\|_2$ Cannot Separate (I_1, I_2)
True action invariant I_2
True action invariant I_1
Single total radius $\|\bar{\rho}\|_2$
d
1-Scalar vs. 2-Block Invariant Readout R^2 (5 Seeds)
1D Projection Ceiling (~0.50)
1-Scalar Readout ($\|z\|_2 \in \mathbf{R}$)
2-Block Readout ($(\bar{r}_1, \bar{r}_2) \in \mathbf{R}^2$)
Held-out (I_1, I_2) Readout R^2
2-Block Geodesic ($K = 2$)
Single-Sphere ($K = 1$)
Unconstrained AE ($d_z = 4$)